



Data Converters

Chapter 01

Sampling, Quantization, & Reconstruction

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Acknowledgement

Concepts & slides are partially based on the lecture

VLSI Data Conversion Circuits

held by

Prof. B. Murmann, PhD, University of Hawai'i

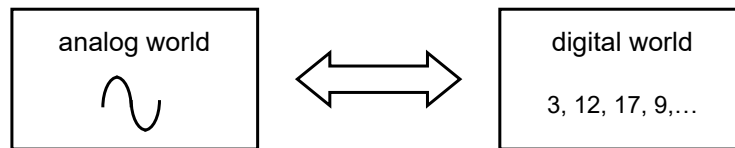
during his sabbatical at the University of Freiburg in 2009



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Our Task – Data Conversion



- real world signals
 - continuous time, continuous amplitude
- digital abstraction
 - discrete time, discrete amplitude
- two tasks
 - discretize in time and amplitude → A/D conversion
 - “undiscretize” in time and amplitude → D/A conversion

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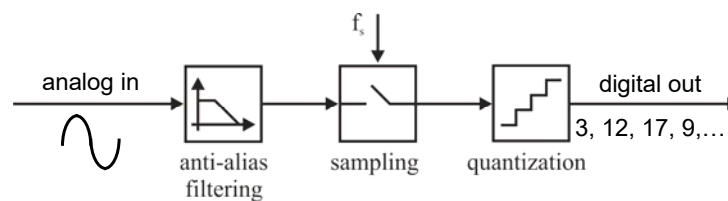
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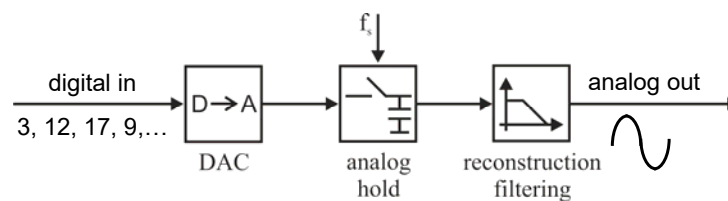
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Overview

A/D conversion



D/A conversion



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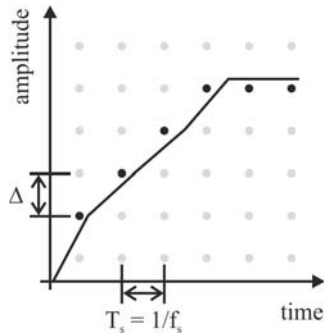
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Uniform Sampling & Quantization



- common approach
 - uniformly sampling in time
 - uniformly quantizing in amplitude
- key questions
 - How fast should sampling be performed?
→ must avoid “aliasing”
 - How much “noise” is added due to amplitude quantization?
 - How to reconstruct the signal back into analog form?

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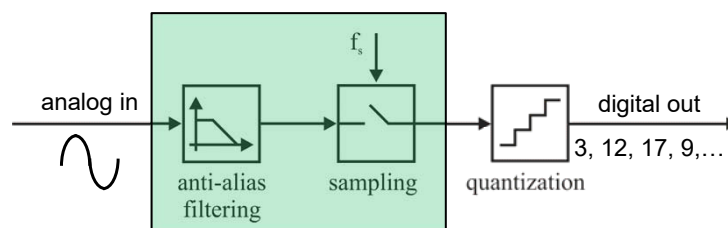
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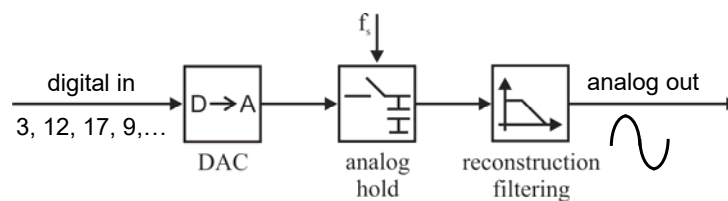
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Overview – Next Topic

A/D conversion



D/A conversion



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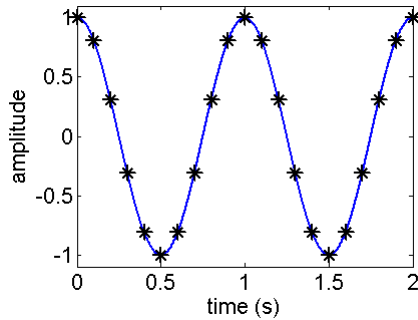
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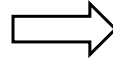
Aliasing – Example I



$$f_{sig} = 1\text{Hz}$$

$$f_s = \frac{1}{T_s} = 10\text{Hz}$$

$$v(t) = \cos(2\pi f_{sig} t)$$



$$t \rightarrow n T_s = \frac{n}{f_s}$$

$$v[n] = \cos[2\pi f_{sig} n T_s]$$

$$= \cos\left[2\pi \frac{1}{10} n\right]$$

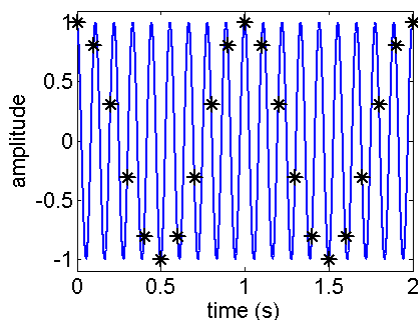
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Aliasing – Example II

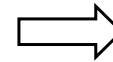


$$f_{sig} = 1\text{Hz}$$

$$f_{sig,2} = 9\text{Hz} = f_s - f_{sig}$$

$$f_s = \frac{1}{T_s} = 10\text{Hz}$$

$$v(t) = \cos(2\pi f_{sig,2} t)$$



$$t \rightarrow n T_s = \frac{n}{f_s}$$

$$v[n] = \cos[2\pi f_{sig,2} n T_s]$$

$$= \cos\left[2\pi \frac{9}{10} n\right]$$

$$= \cos\left[2\pi n - 2\pi \frac{1}{10} n\right]$$

$$= \cos\left[2\pi \frac{1}{10} n\right]$$

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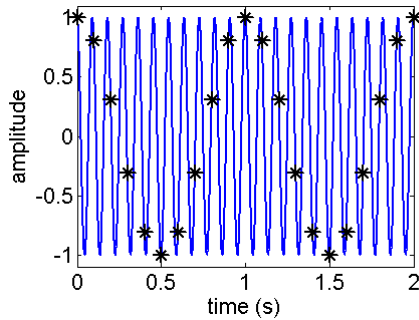
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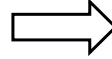
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Aliasing – Example III



$$v(t) = \cos(2\pi f_{sig,3} t)$$



$$t \rightarrow n T_s = \frac{n}{f_s}$$

$$v[n] = \cos[2\pi f_{sig,2} n T_s]$$

$$= \cos\left[2\pi \frac{11}{10} n\right]$$

$$= \cos\left[2\pi n + 2\pi \frac{1}{10} n\right]$$

$$= \cos\left[2\pi \frac{1}{10} n\right]$$

$$f_{sig} = 1\text{Hz}$$

$$f_{sig,3} = 11\text{Hz} = f_s + f_{sig}$$

$$f_s = \frac{1}{T_s} = 10\text{Hz}$$

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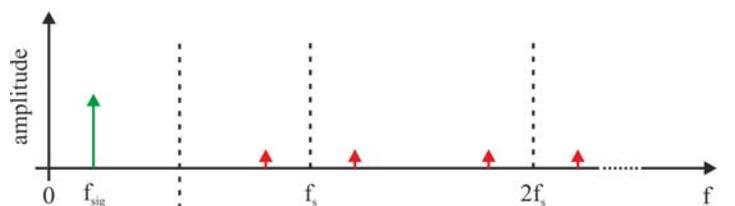
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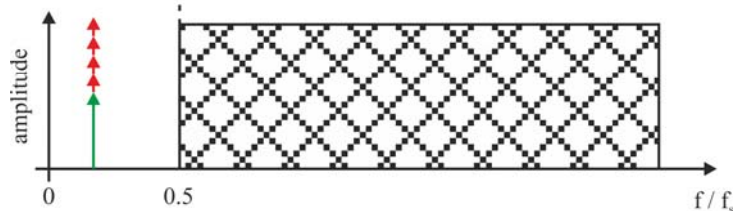
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Consequence

continuous time



discrete time



The frequencies f_{sig} and $n \cdot f_s \pm f_{sig}$ (n integer) are indistinguishable in the discrete-time domain.

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Sampling Theorem

- in order to prevent aliasing

$$f_{\text{sig,max}} = \frac{f_s}{2}$$

- the sampling rate $f_s = 2f_{\text{sig,max}}$ is called the Nyquist sampling rate
- two possibilities
 - sample fast enough to cover all spectral components, including “parasitic” ones outside band of interest
 - limit $f_{\text{sig,max}}$ through filtering → **anti-aliasing filter**

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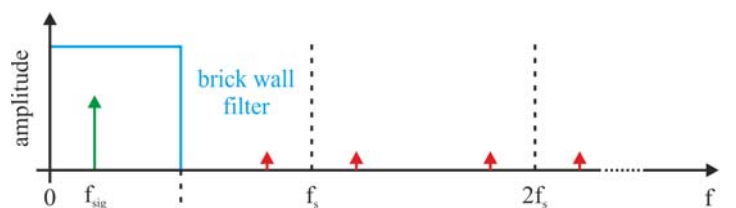
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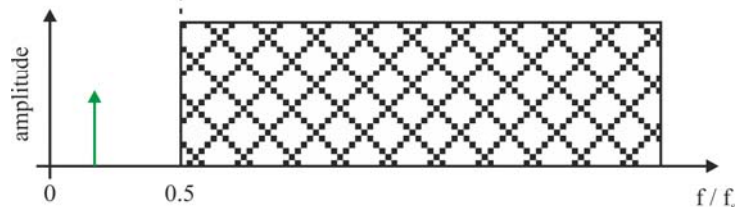
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Perfect Anti-Aliasing Filter – Brick Wall

continuous time



discrete time



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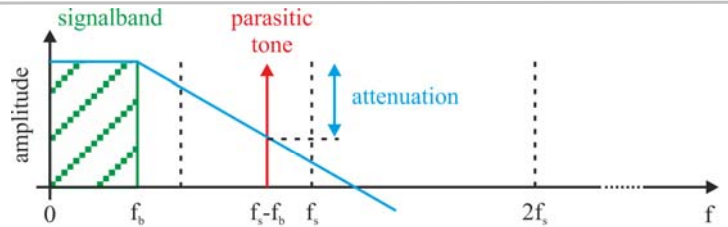
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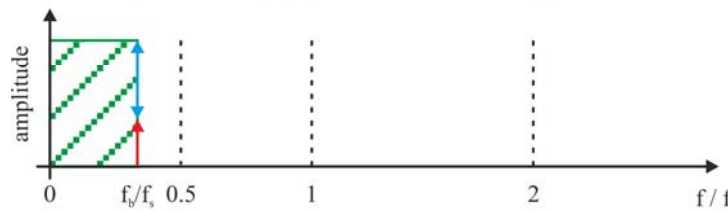


Real Anti-Aliasing Filter

continuous time



discrete time



Sampling rate faster than Nyquist required for adequate attenuation!

→ "oversampling"

→ tradeoff sampling rate ↔ filter order

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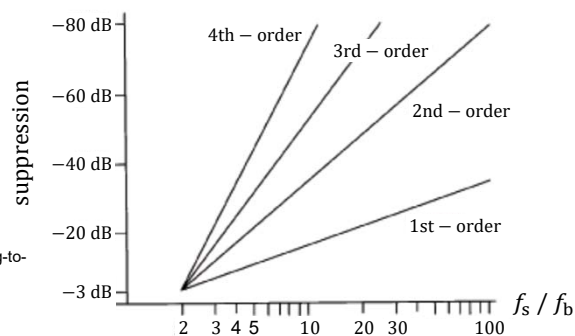
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Tradeoff Sampling Rate ↔ Filter Order



R. van der Plassche, "CMOS Integrated Analog-to-Digital and Digital-to-Analog Converters," Kluwer Academic Publishers, page 41

- low- to medium-speed converters: tradeoff sampling rate ↔ filter order
- high-speed converters: $f_s/f_b > 10$ impossible or too costly to implement
→ high-order filter required

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Classes of Sampling

- **Nyquist data converters**
 - Nyquist-rate sampling $f_s > 2f_{\text{sig,max}}$
 - in practice always slightly oversampled
- **Oversampled data converters**
 - oversampling $f_s \gg 2f_{\text{sig,max}}$
 - anti-aliasing filter may easily be implemented
 - oversampling also reduces “quantization noise” → to be discussed later

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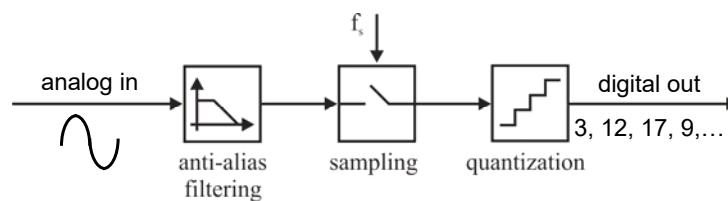
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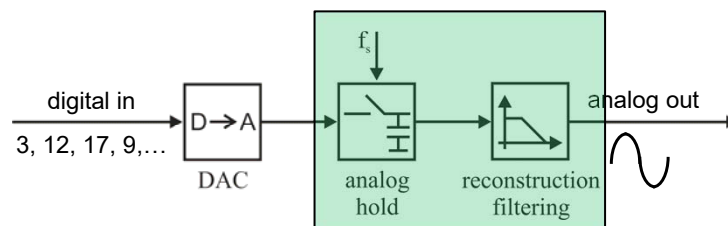
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Overview – Next Topic

A/D conversion



D/A conversion



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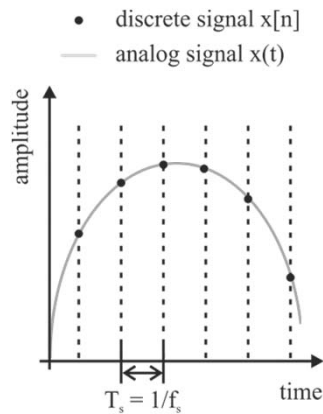
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Reconstruction - Ideal



- How to reconstruct $x(t)$ from $x(n)$?
- $x[n]$ contains all information of $x(t)$ if

$$f_s > 2f_{\text{sig,max}}$$
- signal processing: ideal interpolation

$$x(t) = \sum_{n=-\infty}^{\infty} x[nT_s] g(t - nT_s)$$

$$g(t) = \frac{\sin(\pi f_s t)}{\pi f_s t}$$

- very hard to build an analog circuit that does this...

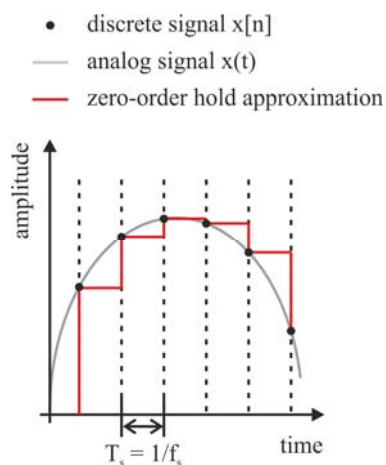
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Reconstruction – Zero-order Hold



- most practical approach
 - simply „hold“ the discrete-time values
 - either for a full period T_s or a fraction thereof
- Effect on the spectrum will be analyzed in two steps on the following slides
 - first: infinitely narrow pulses (Dirac)
 - second: zero-order hold pulses

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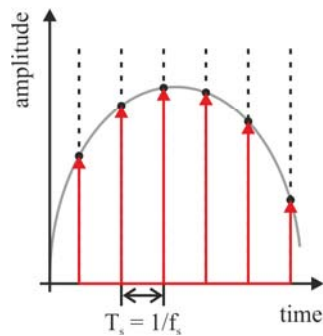
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Reconstruction – Dirac Pulses, Time Domain

- discrete signal $x[n]$
- analog signal $x(t)$
- Dirac pulse signal $x_d(t)$



- time domain

$$x_d(t) = \sum_{n=-\infty}^{\infty} x[nT_s] \delta(t - nT_s)$$

$$= x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

- Multiplication in time domain means convolution in frequency domain!

$$X_d(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X\left(f - \frac{n}{T_s}\right)$$

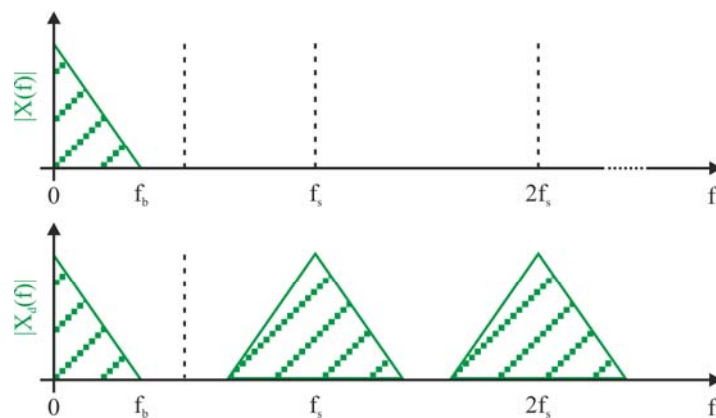
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Reconstruction – Dirac Pulses, Frequency Domain



Spectrum $|X(f)|$ is replicated at integer multiples of the sampling frequency!

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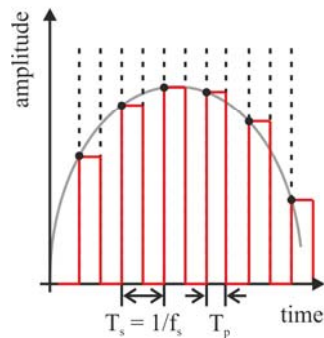
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Reconstruction – Finite Hold Pulses

- discrete signal $x[n]$
- analog signal $x(t)$
- finite hold approximation



- general case: $0 < T_p \leq T_s$
- time domain signal results from convolving the Dirac sequence with a rectangular unit pulse
- spectrum results from multiplication with Fourier transform of the pulse $H_p(f)$

$$H_p(f) = T_p \frac{\sin(\pi f T_p)}{\pi f T_p} e^{-j\pi f T_p}$$

$$X_p(f) = \underbrace{\frac{T_p}{T_s} \frac{\sin(\pi f T_p)}{\pi f T_p}}_{\text{amplitude envelope}} e^{-j\pi f T_p} \sum_{n=-\infty}^{\infty} X\left(f - \frac{n}{T_s}\right)$$

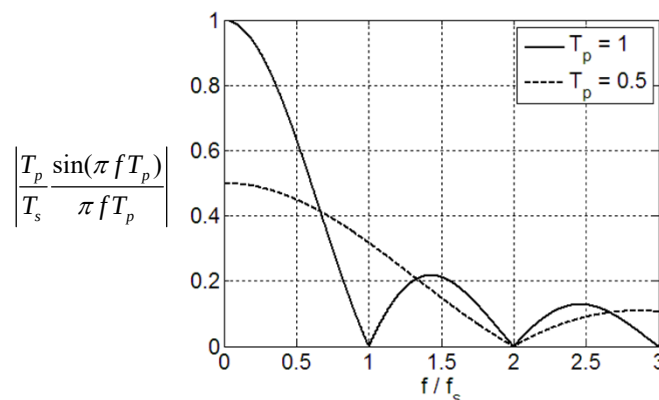
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Reconstruction – Amplitude Envelope



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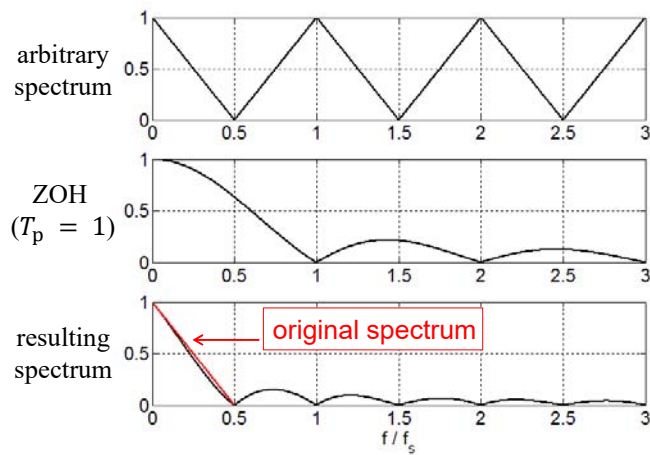
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Example



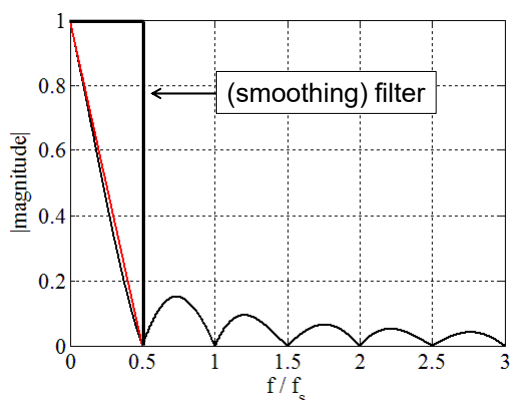
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Reconstruction Filter



as with anti-alias filter:

- brick wall filter would be perfect, but its implementation is too costly... if possible at all...
- oversampling reduces filter order

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First Summary

- The sampling theorem must be met, i.e., $f_s \geq 2 f_{\text{sig,max}}$.
→ anti-aliasing filter may be needed
- If the sampling theorem is met, the continuous-time signal can be recovered from the discrete-time sequence without loss of information.
- typical reconstruction: zero-order hold with smoothing filter
→ pre- or post-emphasis may be needed to cancel drop due to sinc envelope
- oversampling → reduces order of anti-aliasing & reconstruction filter

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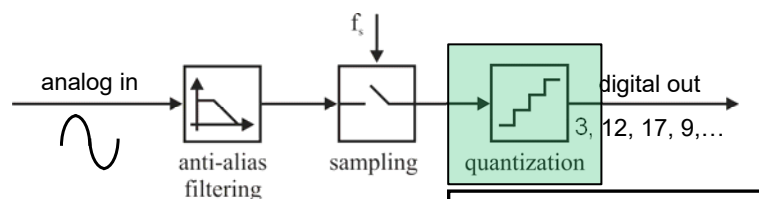
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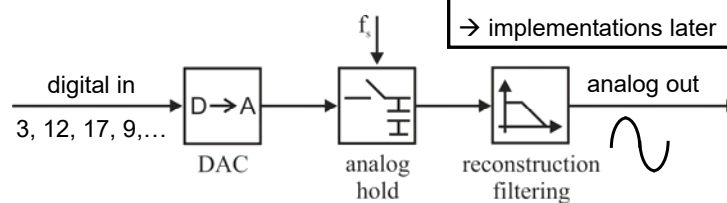
Overview – Next Topic

A/D conversion



transfer function of quantizer
impact of quantization error
→ implementations later

D/A conversion



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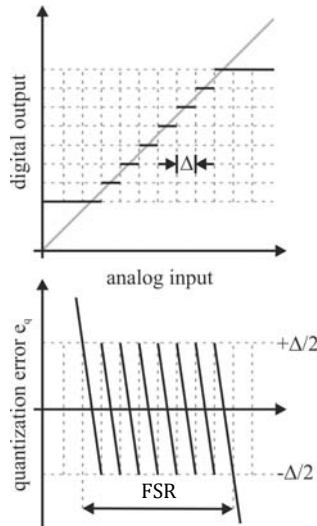
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Quantization of Analog Signals



- quantization step width Δ , also called the *least significant bit LSB*
- quantization error e_q bounded by

$$-\Delta/2 \leq e_q \leq +\Delta/2$$

as long as the analog input signal is within the full-scale range *FSR*

→ quantization error grows out of bounds beyond the full-scale range

→ defined as overloading, i.e., analog input signal is outside the full-scale range *FSR*

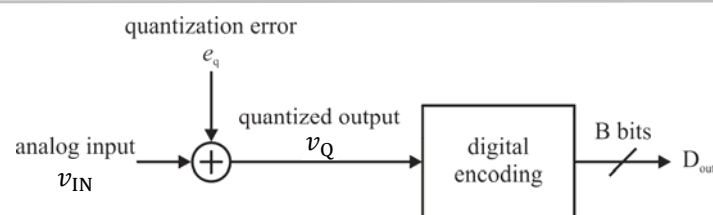
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Conceptual Model of a Quantizer



- digital encoding block determines how quantized levels are mapped into digital codes
 - model does not represent an actual hardware implementation
- shows that quantization and encoding are conceptually separate operations
- changing the encoding has no interesting implications on the function or performance of the quantizer

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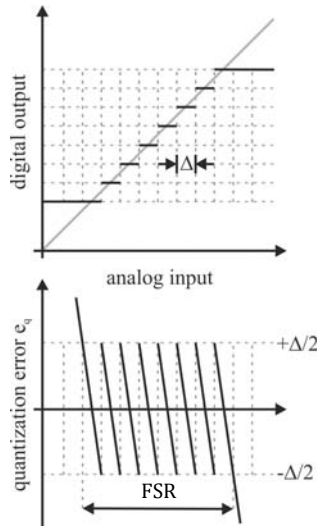
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Encoding Example for a B-Bit Quantizer



- binary encoded number of distinguishable quantization levels determines the resolution of the quantizer
 - here: 8 levels
 - 2^3 levels
 - 3-bit quantizer

$$LSB = \Delta = \frac{FSR}{2^B}$$

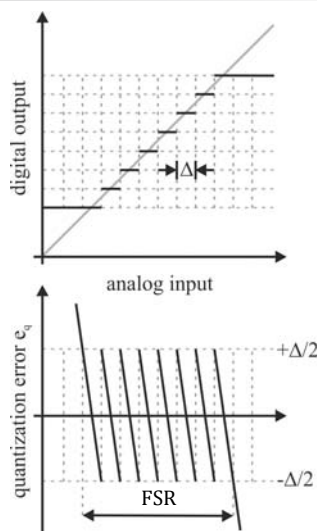
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Encoding Example for a B-Bit Quantizer



- output is a digital word (not an analog voltage)
- typical digital encoding schemes

<u>binary code</u>	<u>thermometer code</u>
111	11111111
110	01111111
101	00111111
⋮	⋮
010	00000011
001	00000001
000	00000000

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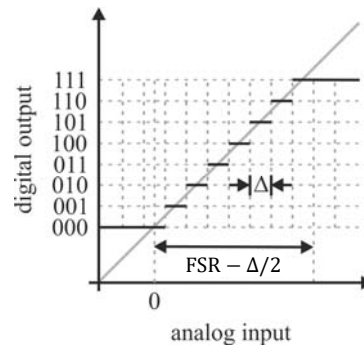
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Types of Quantizers – Unipolar

- unipolar input
→ define origin where first code and straight line fit intersect
- full scale FSR is reduced by $\Delta/2$ (below zero)



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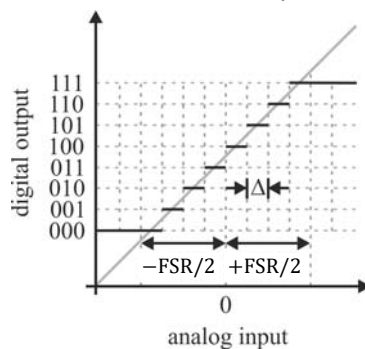
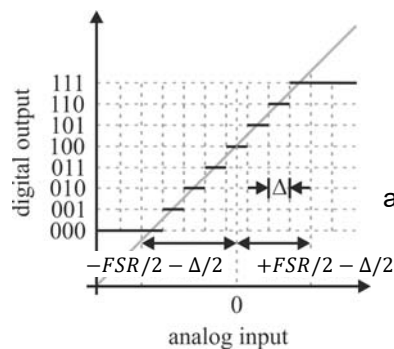
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Types of Quantizers – Bipolar Mid-rise / Mid-tread

- bipolar input
→ quantizer processes negative & positive inputs
→ common case of a differential circuit
- mid-rise / mid-tread
→ zero input coincides with a transition level or not

bipolar
mid-risebipolar
mid-tread
asymmetric FS!

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Effect of Quantization Error on Signal

- two aspects
 - How much power does quantization error add to samples?
 - How is the power distributed in frequency?
- Quantization error is a deterministic function of the signal!
 - We should be able to answer above questions applying deterministic analysis.
 - Unfortunately, such an analysis strongly depends on the signal and may thus become very complex...
- strategy
 - build basic intuition using simple deterministic signals
 - next, abandon basic intuition and revert to a "general" statistical model

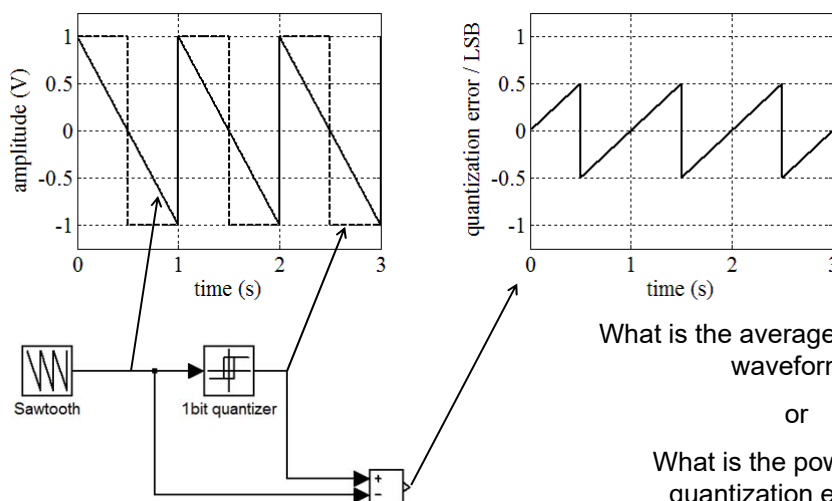
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Example I – Saw-tooth Waveform



What is the average power of this waveform?

or

What is the power of the quantization error e_q ?

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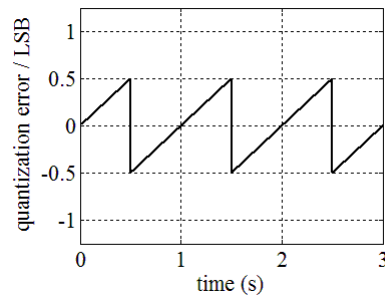
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Example I – Saw-tooth Waveform



$$e_q(t) = \begin{cases} \frac{\Delta}{T}t & 0 \leq t \leq \frac{T}{2} \\ \frac{\Delta}{T}t - \Delta & \frac{T}{2} \leq t \leq T \end{cases}$$

Average power? → Integrate over one period! → mean-square value

$$\overline{e_q^2} = \frac{1}{T} \int_0^T e_q^2(t) dt \longrightarrow \boxed{\overline{e_q^2} = \frac{\Delta^2}{12}}$$

This calculation was no challenge...

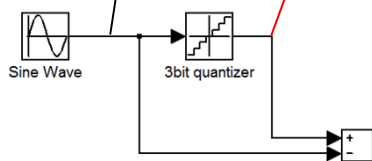
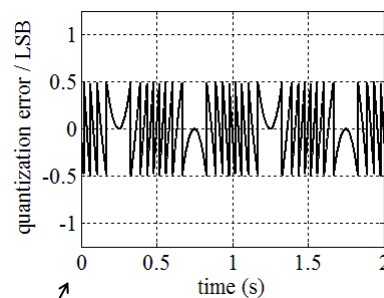
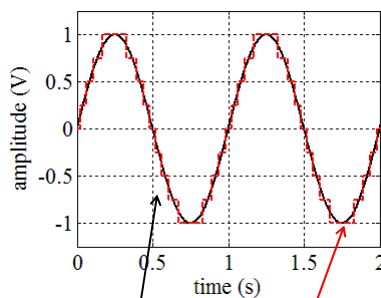
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Example II – Sine Wave



This calculation would be quite a challenge...

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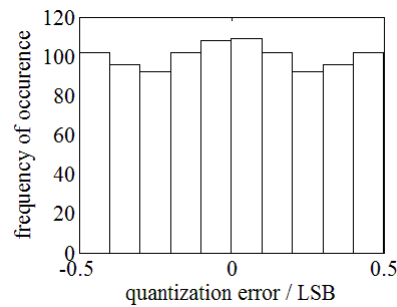
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Quantization Error - Histogram

- sinusoidal input signal with $f_{\text{sig}} = 101$ Hz, sampled at $f_s = 1000$ Hz
- 8-bit quantizer



mean ≈ 0 LSB
variance = $0.982 \text{ LSB}^2/12$

- distribution is “almost” uniform
→ approximate average power by integrating uniform distribution

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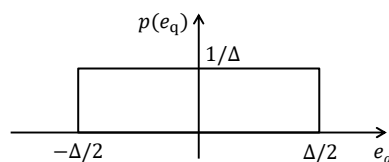
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Quantization Error – Statistical Model

- assumption: e_q has a uniform probability density
- This approximation holds reasonably well in practice if
 - signal covers large number of quantization steps
 - signal is “sufficiently” active
 - quantizer does not overload



mean value $\overline{e_q} = \int_{-\Delta/2}^{+\Delta/2} p(e_q) e_q de_q = 0$

variance $\overline{e_q^2} = \int_{-\Delta/2}^{+\Delta/2} p(e_q) e_q^2 de_q = \frac{\Delta^2}{12}$

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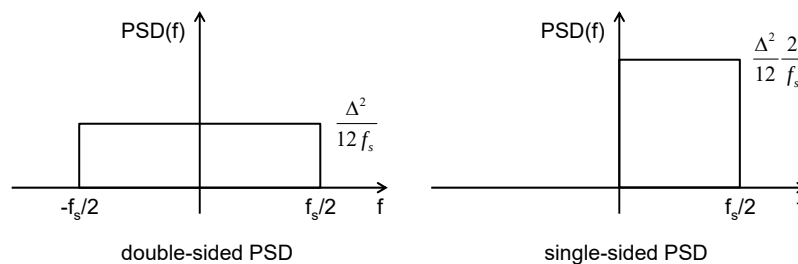
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Quantization Noise Power

It can be shown that the quantization noise power is distributed (approximately) uniformly in frequency.

→ Power Spectral Density (PSD) plot



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Signal-to-Quantization-Noise Ratio – SQNR

Assuming a uniform distribution of e_q and a full-scale sinusoidal input, we get

$$SQNR = \frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2}{12}} = \frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2}{12}} = \frac{3}{2} 2^{2B}$$

$$SQNR = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{FSR}{2} \right)^2}{\frac{\Delta^2}{12}} \right) = 10 \log_{10} \left(\frac{\frac{1}{2} \left(\frac{2^B \Delta}{2} \right)^2}{\frac{\Delta^2}{12}} \right)$$

$$\boxed{SQNR = 6.02 \text{ dB} \times B + 1.76 \text{ dB}}$$

→ Each bit B of the quantizer increases the resolution by 6 dB.

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Data Converters

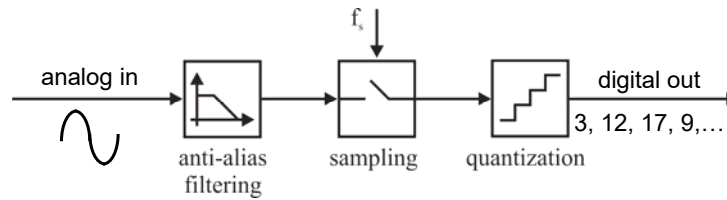
40

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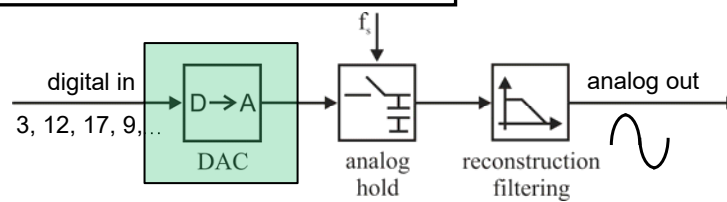
Overview – Next Topic

A/D conversion



transfer function of DAC
→ implementations will be discussed later

D/A conversion



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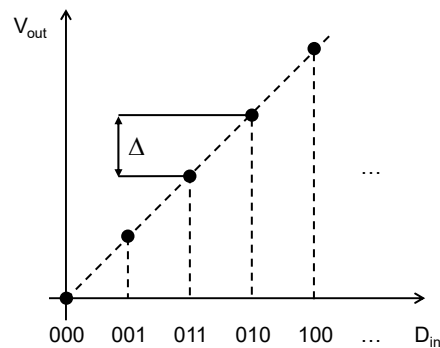
Data Converters

41

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Ideal Digital-to-Analog Converter - DAC

- essentially a digitally controlled voltage, current, or charge source
 - no quantization error!



example: unipolar DAC

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Data Converters

42

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